Key Findings

Phase 1 — How Image Degradation Affects Stereo SLAM
EECE5554 Final Project

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Background

We tested five types of image degradation on a stereo-inertial ORB-SLAM3 system running on two TUM-VI room sequences (room1 and room2). Each degradation was applied to both camera images, simulating real-world lens or sensor degradation.

Two fixed reference points were used per trajectory:

Trajectory	**Clean stereo ATE**	**Clean mono ATE**
Dataset 1 (room1)	0.0068 m	0.0102 m
Dataset 2 (room2)	0.0057 m	0.0082 m

Table 1: Clean baselines for both trajectories.

The mono baseline is the *switching target* — when stereo performance degrades to this level, it is no longer worth running stereo over mono.

Finding 1 — Each Degradation Type Has a Unique Failure Signature

Dataset 1 (room1)

Type	**Parameter**	**Failure at**	**ATE just before cliff**	**Shape**
Gaussian blur	sigma (σ)	$\sigma = 6.1$	0.0102 m (+50%)	Gradual
Motion blur	kernel length	28 pixels	0.0170 m (+150%)	Steep
Salt & pepper	pixel density	40%	0.0074 m (+9%)	Flat then cliff
Brightness	scale factor α	$\alpha = 0.03$	0.0098 m (+44%)	Asymmetric
Occlusion	fraction blocked	61%	0.0086 m (+26%)	Flat then cliff

Table 2: Failure thresholds and degradation shape for room1.

Dataset 2 (room2)

Type	Parameter	Failure at	ATE just before cliff	Shape
Gaussian blur	sigma (σ)	$\sigma = 6.0$	0.0075 m (+31%)	Gradual
Motion blur	kernel length	27 pixels	0.0085 m (+49%)	Steep
Salt & pepper	pixel density	40%	0.0074 m (+30%)	Flat then cliff
Brightness	scale factor α	$\alpha = 0.02$	0.0063 m (+10%)	Asymmetric
Occlusion	fraction blocked	61%	0.0073 m (+28%)	Flat then cliff

Table 3: Failure thresholds and degradation shape for room2.

Not all degradations are equally dangerous or equally detectable. Failure thresholds are scene-dependent but the overall pattern is consistent across both datasets.

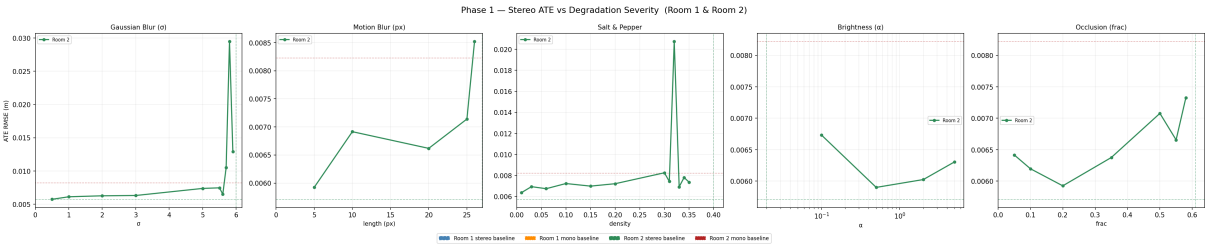

Figure 1: Combined Phase 1 sweep results for all 5 degradation types across Room 1 (blue) and Room 2 (green). Dashed lines = stereo baselines; dotted lines = mono baselines. Vertical dashed lines mark the failure cliff for each trajectory.

Finding 2 — The Mono Baseline is a Natural Switching Point

As degradation increases, stereo ATE rises until it meets the mono baseline — and then SLAM fails. This is not a coincidence.

Stereo SLAM's advantage over mono comes from metric depth via the stereo baseline. As cameras degrade, stereo matching becomes unreliable, the depth anchor is lost, and the system effectively falls back to behaving like mono. One more step of degradation and tracking fails entirely.

The crossover point is therefore both a natural and physically meaningful switching threshold: below it, stereo is earning its advantage; at it, stereo has lost its advantage; beyond it, stereo has failed outright.

Dataset 1 (room1)

- Gaussian blur: crossover at $\sigma = 6.0$ (ATE = 0.0102 m, exactly at mono baseline)
- Motion blur: crossover at length = 25 px (ATE = 0.0170 m, well above mono baseline)

Dataset 2 (room2)

- Gaussian blur: crossover at $\sigma = 5.7$ (ATE = 0.0105 m, just above mono baseline)
- Motion blur: crossover at length = 26 px (ATE = 0.0085 m, just above mono baseline)

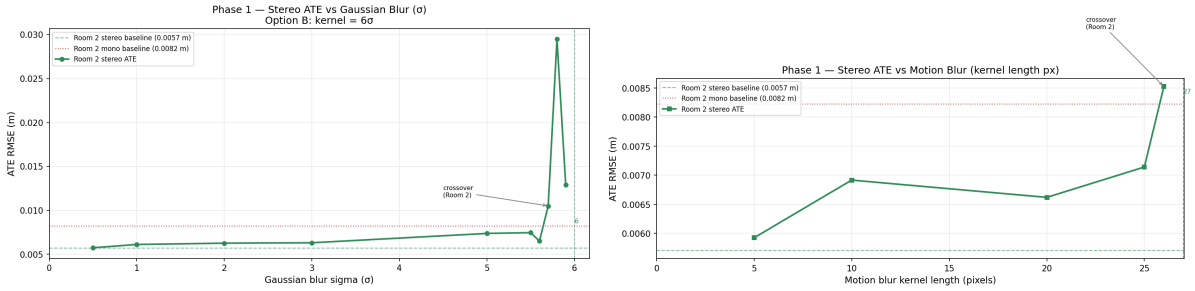

Figure 2: Gaussian blur (left) and motion blur (right) — gradual ATE curves with visible crossover before failure.

This validates the health monitor's severity metric:

$$\text{severity} = \frac{\text{RTE}_{\text{stereo}}}{\text{RTE}_{\text{stereo}} + \text{RTE}_{\text{mono}}}$$

The crossover at severity = 0.5 is the correct trigger point.

Finding 3 — ORB-SLAM3 is Robust to Random Pixel Corruption

Salt & pepper noise and occlusion both showed a **flat ATE curve** across most of their range:

Dataset 1 (room1)

- Salt & pepper: ATE barely changes from 1% up to 35% corrupted pixels
- Occlusion: ATE barely changes from 5% up to 58% of image blocked

Dataset 2 (room2)

- Salt & pepper: ATE stays 0.0064–0.0083 m from 1% up to 35% corrupted pixels
- Occlusion: ATE stays 0.0059–0.0073 m from 5% up to 58% of image blocked

The reason is ORB-SLAM3's built-in outlier rejection (RANSAC). Random corrupted pixels or a contiguous blocked region look like outlier keypoints, which RANSAC discards automatically. SLAM only fails when so much of the image is corrupted that there are not enough good keypoints left to track.

Implication: These degradations produce no learnable gradient for the health monitor. The signal is binary (fine, then suddenly failed) with no warning zone.

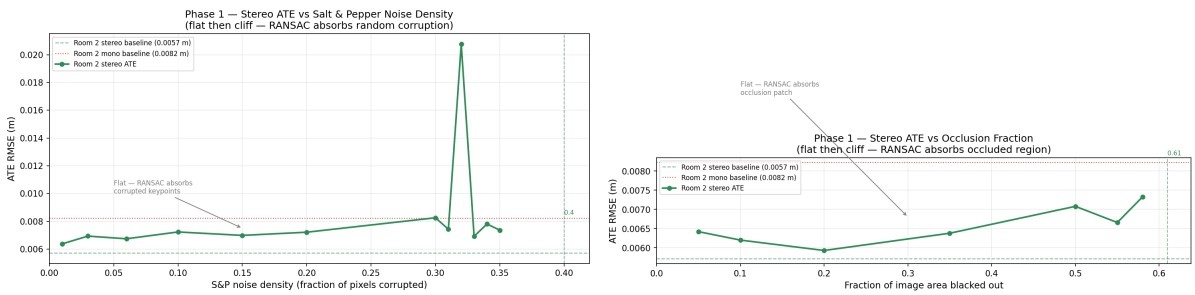

Figure 3: Salt & pepper (left) and occlusion (right) — flat ATE curves followed by a hard cliff.

Finding 4 — Underexposure is Far More Dangerous Than Overexposure

Brightness degradation is strongly asymmetric:

Dataset 1 (room1)

- **Overexposure** up to 10× normal brightness: SLAM survives, ATE only 22% worse
- **Underexposure** below $\alpha = 0.03$: SLAM fails immediately

Dataset 2 (room2)

- **Overexposure** up to 5× normal brightness: SLAM survives, ATE only 10% worse
- **Underexposure** below $\alpha = 0.02$: SLAM fails immediately

The reason: ORB feature detection relies on intensity gradients (edges and corners). In very dark images, all pixel values collapse toward zero and gradients disappear. In bright (overexposed) images, gradients still exist at object boundaries even when highlights are clipped.

Implication: A simple brightness check (mean pixel intensity below a threshold) could detect extreme underexposure without needing a neural network.

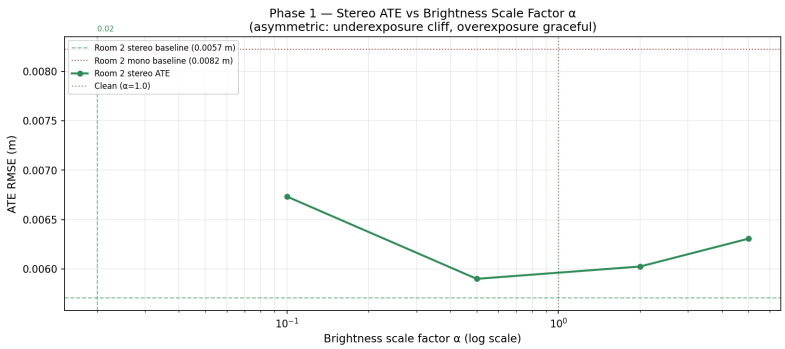

Figure 4: Brightness sweep — asymmetric response: overexposure is benign, underexposure causes immediate failure.

Finding 5 — Gaussian Blur and Motion Blur are the Best Candidates for Health Monitoring

These two degradations produce a **gradual, learnable severity curve**:

Dataset 1 (room1)

Gaussian blur:

- ATE rises steadily from clean (+1.5%) to crossover (+50%) over 7 sigma levels
- Long warning window before failure at $\sigma = 6.1$

Motion blur:

- ATE rises steeply — +150% just before failure at length = 28 px
- Shorter warning window but very strong signal

Dataset 2 (room2)

Gaussian blur:

- ATE rises from clean (+1%) to crossover (+84%) across 7 sigma levels
- Cliff is slightly earlier ($\sigma = 6.0$) than room1, reflecting scene texture differences

Motion blur:

- ATE rises from clean to crossover at length = 26 px (just above mono baseline)
- Cliff is one pixel step later than room1 (27 vs 28 px), consistent behavior

Both types give the health monitor a continuous input-output relationship to learn: as images get blurrier, severity increases monotonically. This is exactly what a regression model needs.

Finding 6 — The Cliff is Sharp and Scene-Dependent

Across all degradation types, once SLAM fails at a given severity level, every higher level also fails. There is no recovery.

Dataset 1 (room1)

The transition zone is typically one parameter step wide: $\sigma = 6.0$ succeeds, $\sigma = 6.1$ fails.

Dataset 2 (room2)

Similarly sharp transitions: $\sigma = 5.9$ succeeds, $\sigma = 6.0$ fails. Motion blur fails at length = 27 px vs. room1's 28 px.

The exact threshold varies by scene (texture density, lighting, scene depth), but the cliff pattern is consistent: success up to a hard limit, then total failure.

This has two implications:

1. The health monitor must act *before* the cliff, not at it
2. There is no benefit to running stereo once the cliff is crossed — immediate switch to mono is the right policy

Generalised Inference

The following conclusions hold across both datasets:

1. **Blur (Gaussian and motion) is the most informative degradation type.** Both produce a monotonically increasing ATE curve that crosses the mono baseline before SLAM fails — giving the health monitor a clear, learnable signal with a meaningful warning window.
2. **Random corruption (S&P, occlusion) is absorbed by RANSAC.** ATE stays flat across a wide severity range, then collapses in one step. No warning zone exists, making these poor health monitor targets.
3. **Brightness has an asymmetric response.** Overexposure is benign; underexposure is catastrophic. A simple rule-based check is sufficient — no learned model needed.
4. **Failure thresholds are scene-dependent but patterns are not.** Room2 fails at slightly lower blur than room1, but both show the same crossover-then-cliff structure. A

model trained across multiple scenes will generalise better than one trained on a single trajectory.

5. **The mono baseline crossover is a universal switching anchor.** In both trajectories, the point where stereo ATE equals mono ATE corresponds to the last stable operating point before SLAM failure. This makes severity = 0.5 a physically grounded and scene-agnostic trigger threshold.

Recommendation for Phase 2

Based on these findings, Phase 2 (health monitor training) should:

1. **Focus training data on Gaussian blur and motion blur** — these provide the richest severity gradients across both trajectories
2. **Use data from both trajectories** — room1 and room2 differ in texture and lighting, increasing model generalisation
3. **Use the mono baseline crossover as the severity=1 anchor** — already implemented via the ratio policy
4. **Target early detection** — the model should predict high severity one or two steps before the cliff, not at the cliff itself
5. **Consider salt & pepper and occlusion as hard negatives** — the model should learn to output low severity for these even at high degradation levels, since SLAM is unaffected